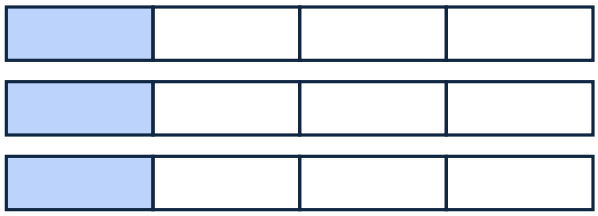
Fractions as Division

Share wholes equally: each share is a fraction.

3 pizzas shared by 4 kids



Cut every pizza into 4 pieces.
Each kid gets 1 piece of each pizza:
3 pieces of 1/4. That is 3/4.

$$3 \div 4 = \frac{3}{4}$$

The top is shared by the bottom

$$2 \div 5 = \frac{2}{5} \qquad 5 \div 2 = \frac{5}{2}$$

The fraction bar means "shared by".

More than one each: 7 ÷ 3

$$7 \div 3 = \frac{7}{3} = 2\,\frac{1}{3}$$

Each kid gets 2 whole ones,
then the last one is cut in 3.

Your turn!

$$4 \div 5 = \frac{4}{5}$$

For teachers

Goal: a fraction is a division (5.NF.3). Share real things: 3 pizzas among 4 kids. Say "How many wholes? How many people? Each gets top over bottom."

Fractions as Division: Practice

The top is shared by the bottom.

Part 1 Share the strips

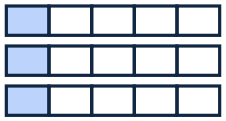
One share is shaded. Count its pieces.

1) 2 shared by 3



$2 \div 3 = \frac{2}{3}$

2) 3 shared by 5



$3 \div 5 = \frac{3}{5}$

3) 4 shared by 6



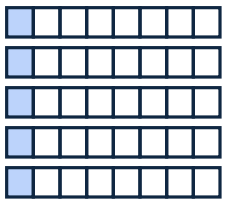
$4 \div 6 = \frac{4}{6}$

4) 2 shared by 5



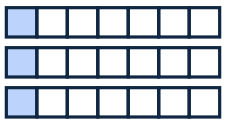
$2 \div 5 = \frac{2}{5}$

5) 5 shared by 8



$5 \div 8 = \frac{5}{8}$

6) 3 shared by 7



$3 \div 7 = \frac{3}{7}$

Part 2 Write it as a fraction

No help this time.

7) $5 \div 6 = \frac{5}{6}$

8) $2 \div 9 = \frac{2}{9}$

9) $7 \div 8 = \frac{7}{8}$

10) $1 \div 4 = \frac{1}{4}$

11) $4 \div 9 = \frac{4}{9}$

12) $3 \div 10 = \frac{3}{10}$

Fractions as Division: Practice (continued)

Part 2 Write it as a fraction (continued)

13) $6 \div 7 = \frac{6}{7}$

14) $5 \div 12 = \frac{5}{12}$

15) $2 \div 3 = \frac{2}{3}$

16) $8 \div 9 = \frac{8}{9}$

17) $1 \div 6 = \frac{1}{6}$

18) $7 \div 10 = \frac{7}{10}$

Part 3 More than one each No help this time. Write the fraction, then the mixed number.

19) $7 \div 2 = \frac{7}{2} = 3\frac{1}{2}$

20) $11 \div 4 = \frac{11}{4} = 2\frac{3}{4}$

21) $9 \div 5 = \frac{9}{5} = 1\frac{4}{5}$

22) $13 \div 3 = \frac{13}{3} = 4\frac{1}{3}$

23) $17 \div 6 = \frac{17}{6} = 2\frac{5}{6}$

24) $10 \div 3 = \frac{10}{3} = 3\frac{1}{3}$

Fractions as Division: Practice (continued)

Part 3 More than one each (continued)

25) $19 \div 4 = \frac{19}{4} = 4\frac{3}{4}$

26) $23 \div 5 = \frac{23}{5} = 4\frac{3}{5}$

27) $15 \div 8 = \frac{15}{8} = 1\frac{7}{8}$

28) $25 \div 6 = \frac{25}{6} = 4\frac{1}{6}$

Part 4 Between which wholes? No help this time.

29) 17 ÷ 5 is between 3 and 4

30) 29 ÷ 4 is between 7 and 8

31) 40 ÷ 7 is between 5 and 6

32) 22 ÷ 3 is between 7 and 8

33) 51 ÷ 8 is between 6 and 7

34) 33 ÷ 6 is between 5 and 6

35) 46 ÷ 9 is between 5 and 6

36) 26 ÷ 10 is between 2 and 3

Fractions as Division: Practice (continued)

Part 5 Who gets more? No help this time. Write < or >.

37) $3 \div 4$ [>] $2 \div 4$

38) $5 \div 6$ [>] $5 \div 8$

39) $2 \div 3$ [<] $4 \div 3$

40) $7 \div 10$ [<] $7 \div 5$

41) $1 \div 2$ [>] $3 \div 8$

42) $4 \div 5$ [<] $9 \div 10$

Part 6 Write it as a fraction

43) $1 \div 11 = \frac{1}{11}$

44) $4 \div 11 = \frac{4}{11}$

45) $2 \div 11 = \frac{2}{11}$

46) $7 \div 9 = \frac{7}{9}$

47) $11 \div 12 = \frac{11}{12}$

48) $10 \div 12 = \frac{10}{12}$

Fractions as Division: Practice (continued)

Part 6 Write it as a fraction (continued)

49) $9 \div 11 = \frac{9}{11}$

50) $10 \div 11 = \frac{10}{11}$

51) $3 \div 11 = \frac{3}{11}$

52) $7 \div 12 = \frac{7}{12}$

53) $2 \div 12 = \frac{2}{12}$

54) $8 \div 10 = \frac{8}{10}$

55) $2 \div 5 = \frac{2}{5}$

56) $1 \div 12 = \frac{1}{12}$

57) $2 \div 6 = \frac{2}{6}$

58) $9 \div 12 = \frac{9}{12}$

59) $3 \div 12 = \frac{3}{12}$

60) $6 \div 11 = \frac{6}{11}$

Fractions as Division: Practice (continued)

Part 6 Write it as a fraction (continued)

61) $3 \div 9 = \frac{3}{9}$

62) $1 \div 10 = \frac{1}{10}$

63) $4 \div 12 = \frac{4}{12}$

64) $2 \div 8 = \frac{2}{8}$

65) $1 \div 2 = \frac{1}{2}$

66) $1 \div 3 = \frac{1}{3}$

Part 7 More than one each

67) $26 \div 7 = \frac{26}{7} = 3\frac{5}{7}$

68) $29 \div 5 = \frac{29}{5} = 5\frac{4}{5}$

69) $23 \div 4 = \frac{23}{4} = 5\frac{3}{4}$

70) $13 \div 2 = \frac{13}{2} = 6\frac{1}{2}$

71) $8 \div 7 = \frac{8}{7} = 1\frac{1}{7}$

72) $35 \div 2 = \frac{35}{2} = 17\frac{1}{2}$

Fractions as Division: Practice (continued)

Part 7 More than one each (continued)

73) $8 \div 3 = \frac{8}{3} = 2\frac{2}{3}$

74) $34 \div 5 = \frac{34}{5} = 6\frac{4}{5}$

75) $39 \div 5 = \frac{39}{5} = 7\frac{4}{5}$

76) $12 \div 7 = \frac{12}{7} = 1\frac{5}{7}$

77) $39 \div 9 = \frac{39}{9} = 4\frac{3}{9}$

78) $26 \div 9 = \frac{26}{9} = 2\frac{8}{9}$

79) $34 \div 7 = \frac{34}{7} = 4\frac{6}{7}$

80) $26 \div 8 = \frac{26}{8} = 3\frac{2}{8}$

81) $11 \div 7 = \frac{11}{7} = 1\frac{4}{7}$

82) $19 \div 8 = \frac{19}{8} = 2\frac{3}{8}$

83) $10 \div 6 = \frac{10}{6} = 1\frac{4}{6}$

84) $22 \div 3 = \frac{22}{3} = 7\frac{1}{3}$

Fractions as Division: Practice (continued)

Part 8 Remember? Problems from earlier packets.

85) $\frac{4}{5} \times \frac{5}{6} = \frac{20}{30}$

86) $\frac{2}{5} + \frac{1}{6} = \frac{17}{30}$

87) $4 \times 3 \times 5 =$ 60

88) $0.8 \times 0.4 =$ 0.32

89) $1.6 \div 4 =$ 0.4

90) $426 \times 13 =$ 5,538

Part 9 Word problems Who shares? How many wholes?

91) 4 friends share 3 sandwiches equally. How much does each friend get?

Number sentence 3 ÷ 4 = 3/4

Answer 3/4 sandwich

Draw or show your work

92) 5 m of ribbon is cut into 8 equal pieces. How long is each piece?

Number sentence 5 ÷ 8 = 5/8

Answer 5/8 m

Draw or show your work

93) 7 cups of juice are poured equally into 2 jugs. How much is in each jug?

Number sentence 7 ÷ 2 = 7/2 = 3 1/2

Answer 3 1/2 cups

Draw or show your work

Fractions as Division: Practice (continued)

Part 9 Word problems (continued)

94) 6 kids share 10 granola bars. How many bars does each kid get?

Number sentence 10 ÷ 6 = 10/6 = 1 4/6

Answer 1 4/6 bars

Draw or show your work

95) Challenge: 3 pies go to 5 kids, and 2 pies go to 3 kids. Which kids get more pie?

Number sentence 3/5 < 2/3, so the 3 kids

Answer the 3 kids

Draw or show your work

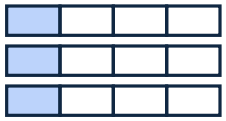
Check-In: Fractions as Division

Show what you know. No help this time.

Mastery: 7 of 8 (80%)

Part 1 Share the strips

1) 3 shared by 4



$3 \div 4 = \frac{3}{4}$

Part 2 Write it as a fraction

2) $9 \div 10 = \frac{9}{10}$

3) $3 \div 8 = \frac{3}{8}$

4) $5 \div 9 = \frac{5}{9}$

Part 3 More than one each

5) $14 \div 3 = \frac{14}{3} = 4\frac{2}{3}$

6) $9 \div 2 = \frac{9}{2} = 4\frac{1}{2}$

Part 4 Between which wholes?

7) 38 ÷ 5 is between 7 and 8

8) 50 ÷ 6 is between 8 and 9